



General Sir John Kotelawala Defence University

Department of Computer Science

CS11021– Programming Laboratory (Intake 43)

Lab Sheet 7

1. Develop a C++ program to determine the type of scholarship a student is eligible for based on their GPA and family income. If the GPA is greater than or equal to 3.7 and the family income is less than 2000, the student should be awarded a Full Scholarship. If the GPA is greater than or equal to 3.5 and the income is less than 4000, award a Partial Scholarship. If the GPA is greater than or equal to 3.0 but does not meet the above conditions, award a Merit Certificate Only. Otherwise, display Not Eligible.
2. Develop a C++ program that calculates the applicable loan interest rate based on the applicant's employment type and credit score. Permanent employees with a credit score of 750 or above should receive an interest rate of 8%, while permanent employees with a score between 650 and 749 should receive a rate of 10%. Contract employees with a credit score of 700 or above should receive a rate of 12%. All other applicants should be assigned an interest rate of 15%.
3. Develop a C++ program to calculate the total charging fee for an electric vehicle based on the charging time and energy consumption. If charging occurs during peak hours (6.00 AM to 10.00 AM or 6.00 PM to 10.00 PM), the cost per kWh is \$0.30, otherwise the off-peak rate of \$0.20 per kWh applies. If the total energy consumption exceeds 50 kWh, an additional 5% service charge should be added to the final bill.
4. Develop a C++ program to validate whether a student is eligible to register for an advanced programming course. The student must have obtained at least 70 marks in Programming, at least 65 marks in Mathematics, and must have an attendance percentage of 80% or above. If the student fails to meet any of these conditions, the program should clearly indicate the reason or reasons for rejection using appropriate conditional statements.

5. Develop a C++ program that calculates the total hotel bill based on room type, season, and number of days stayed. A Deluxe room costs \$200 per day during peak season and \$150 during off-peak season, while a Standard room costs \$120 during peak season and \$90 during off-peak season. If the guest stays for more than five days, a 10% discount should be applied to the total amount.
6. Develop a C++ program to determine system access permissions based on the user's role and login time. An Admin user is allowed access at any time, while Staff members are allowed access only between 08.00 and 18.00 hours. Guest users are permitted access only between 10.00 and 16.00 hours. The program should use a switch-case structure combined with conditional checks to determine access status.
7. Develop a C++ program that evaluates a vehicle's fuel efficiency and environmental impact. If the mileage is 25 km/L or above and the emission level is below 100 g/km, classify the vehicle as *Eco Friendly*. If the mileage is between 15 and 24 km/L, classify it as *Standard Efficiency*. If the mileage is below 15 km/L, classify it as *Low Efficiency*. However, if the emission level exceeds 150 g/km, the vehicle must be classified as a *Polluting Vehicle* regardless of mileage.
8. Develop a C++ program to analyze a student's final examination result based on marks obtained in three subjects. The student must score at least 50 marks in each subject to pass. If the average mark is 60 or above, assign *Second Class*; if it is 75 or above, assign *First Class*; and if it is 85 or above, assign *Distinction*. If the student scores below 50 in any subject, the result should be *Fail*, irrespective of the average.